

Paper Reference(s)

9801/01

Edexcel

Mathematics

Advanced Extension Award

Wednesday 27 June 2018 – Morning

Time: 3 hours

Materials required for examination

Mathematical Formulae (Pink)

Items included with question papers

Answer book (AB16)

Graph paper (ASG2)

Candidates may NOT use a calculator in answering this paper.

Instructions to Candidates

In the boxes on the answer book provided, write the name of the examining body (Edexcel), your centre number, candidate number, the paper title (Mathematics), the paper reference (9801), your surname, initials and signature.

Check that you have the correct question paper.

Answers should be given in as simple a form as possible. e.g. $\frac{2\pi}{3}$, $\sqrt{6}$, $3\sqrt{2}$.

Information for Candidates

A booklet 'Mathematical Formulae and Statistical Tables' is provided.

Full marks may be obtained for answers to ALL questions.

The marks for individual questions and the parts of questions are shown in round brackets: e.g. (2).

There are 7 questions in this question paper.

The total mark for this paper is 100, of which 7 marks are for style, clarity and presentation.

There are 8 pages in this question paper. Any blank pages are indicated.

Advice to Candidates

You must ensure that your answers to parts of questions are clearly labelled.

You must show sufficient working to make your methods clear to the Examiner.

Answers without working may not gain full credit.

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1. (a) Show that $\sqrt{\frac{1+x}{1-x}}$ can be written in the form $\frac{1+x}{\sqrt{1-x^2}}$ for $|x| < 1$ (1)

(b) Hence, or otherwise, find the expansion, in ascending powers of x up to and including the term in x^5 , of $\sqrt{\frac{1+x}{1-x}}$ (4)

(Total 5 marks)

2. Solve, for $0 \leq x \leq 360^\circ$

$$\sin 47^\circ \cos^3 x + \cos 47^\circ \sin x \cos^2 x = \frac{1}{2} \cos^2 x$$

(7)

(Total 7 marks)

3. The lines L_1 and L_2 have the equations

$$L_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 9 \end{pmatrix} + s \begin{pmatrix} 2 \\ p \\ 6 \end{pmatrix} \quad \text{and} \quad L_2 : \mathbf{r} = \begin{pmatrix} -15 \\ 12 \\ -9 \end{pmatrix} + t \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$$

where p is a constant.

The acute angle between L_1 and L_2 is θ where $\cos \theta = \frac{\sqrt{5}}{3}$

(a) Find the value of p . (5)

The line L_3 has equation $\mathbf{r} = \begin{pmatrix} -15 \\ 12 \\ -9 \end{pmatrix} + u \begin{pmatrix} 8 \\ -6 \\ -5 \end{pmatrix}$ and the lines L_3 and L_2 intersect at the point A .

The point B on L_2 has position vector $\begin{pmatrix} 5 \\ -13 \\ 1 \end{pmatrix}$ and point C lies on L_3 such that $ABDC$ is a rhombus.

(b) Find the two possible position vectors of D . (5)

(Total 10 marks)

4. A curve C has equation $y = f(x)$ where $x \in \mathbb{R}$ and f is a one-one function.

- (a) Describe a single transformation that transforms C to the curve with equation $y = -f(-x)$. (1)

The curve C is reflected in the line with equation $y = -x$ to give the curve V .
The equation of V is $y = g(x)$.

- (b) Explain why $g^{-1}(x) = -f(-x)$. (1)

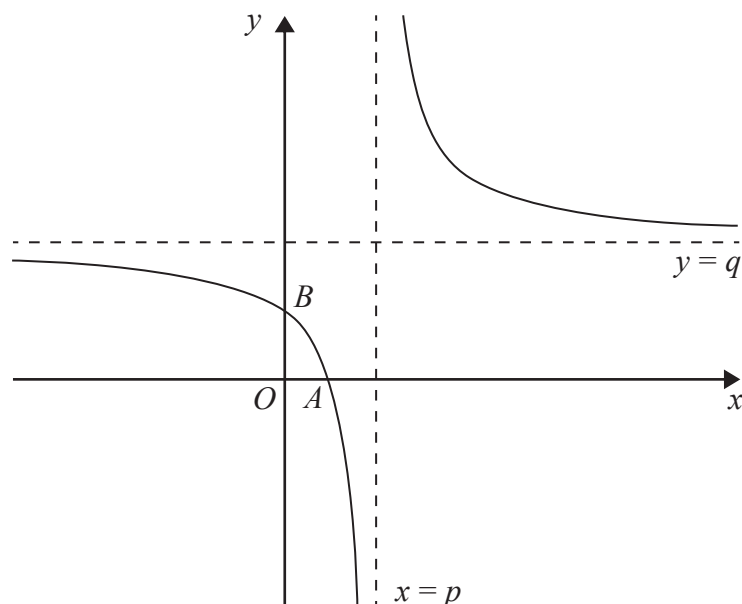


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$ where

$$f(x) = \frac{3(x-1)}{x-2} \quad x \in \mathbb{R}, x \neq 2$$

The curve has asymptotes with equations $x = p$ and $y = q$ and C crosses the x -axis at the point A and the y -axis at the point B .

- (c) Write down the value of p and the value of q . (1)

- (d) Write down the coordinates of the point A and the coordinates of the point B . (1)

Given the definition of $g(x)$ in part (b),

- (e) find the function g . (4)

- (f) Solve $g^{-1}f(x) = x$ (5)

(Total 13 marks)

5.

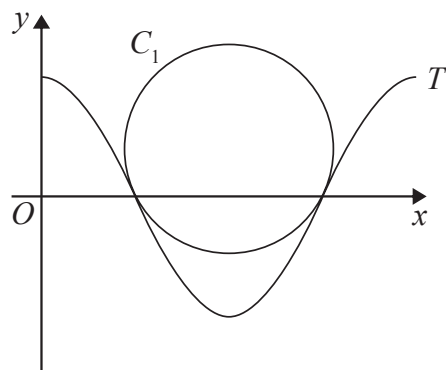


Figure 2

Figure 2 shows part of the curve T with equation $y = \cos 2x$ and the circle C_1 that touches T at $\left(\frac{\pi}{4}, 0\right)$ and $\left(\frac{3\pi}{4}, 0\right)$.

(a) Find the radius of C_1

(5)

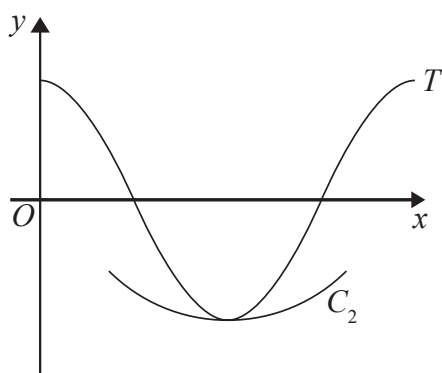


Figure 3

Figure 3 shows a sketch of part of T and part of a circle C_2 that touches T at the point P with coordinates $\left(\frac{\pi}{2}, -1\right)$. For values of x close to $\frac{\pi}{2}$ the curve T lies inside C_2 as shown in Figure 3.

(b) Without doing any calculation, explain why the value of $\frac{d^2y}{dx^2}$ for C_2 at P is less than the value of $\frac{d^2y}{dx^2}$ for T at P .

(1)

The radius of C_2 is r .

(c) Use the result from part (b) to find a value of k such that $r > k$.

(6)

Given that C_2 cuts T at the point $(0, 1)$,

(d) find the value of r .

(2)

(Total 14 marks)

6. (a) Use the substitution $u = \sqrt{t}$ to show that

$$\int_1^x \frac{\ln t}{\sqrt{t}} dt = 4 - 4\sqrt{x} + 2\sqrt{x} \ln x \quad x \geq 1$$

(5)

- (b) The function g is such that

$$\int_1^x g(t) dt = x - \sqrt{x} \ln x - 1 \quad x \geq 1$$

- (i) Use differentiation to find the function g .

- (ii) Evaluate $\int_4^{16} g(t) dt$ and simplify your answer.
- (5)

- (c) Find the value of x (where $x > 1$) that gives the maximum value of

$$\int_x^{x+1} \frac{\ln t}{2^t} dt$$

(7)

(Total 17 marks)

7.

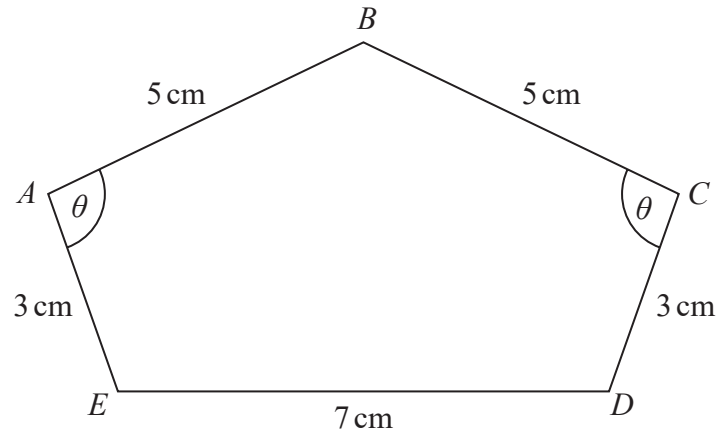


Figure 4

Figure 4 shows a shape $S(\theta)$ made up of five line segments AB , BC , CD , DE and EA . The lengths of the sides are $AB = BC = 5$ cm, $CD = EA = 3$ cm and $DE = 7$ cm. Angle $BAE = \text{angle } BCD = \theta$ radians.

The length of each line segment always remains the same but the value of θ can be varied so that different symmetrical shapes can be formed, with the added restriction that none of the line segments cross.

- (a) Sketch $S(\pi)$, labelling the vertices clearly. (2)

The shape $S(\phi)$ is a trapezium.

- (b) Sketch $S(\phi)$ and calculate the value of ϕ . (3)

The smallest possible value for θ is α , where $\alpha > 0$, and the largest possible value for θ is β , where $\beta > \pi$.

- (c) Show that $\alpha = \arccos\left(\frac{29}{40}\right)$. [$\arccos(x)$ is an alternative notation for $\cos^{-1}(x)$] (4)

- (d) Find the value of β . (4)

The area, in cm^2 , of shape $S(\theta)$ is $R(\theta)$.

- (e) Show that for $\alpha \leq \theta < \pi$

$$R(\theta) = 15 \sin \theta + \frac{7}{4} \sqrt{87 - 120 \cos \theta} \quad (4)$$

Given that this formula for $R(\theta)$ holds for $\alpha \leq \theta \leq \beta$

(f) show that $R(\theta)$ has only one stationary point and that this occurs when $\theta = \frac{2\pi}{3}$ (7)

(g) find the maximum and minimum values of $R(\theta)$. (3)

(Total 27 marks)

FOR STYLE, CLARITY AND PRESENTATION: 7 MARKS

TOTAL FOR PAPER: 100 MARKS

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